

EOIS

Executive Office Intelligence System

External Technology Integrations — Reference Document

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Classification: Internal / Confidential

! IMPORTANT NOTICE

The technology choices documented in this report were explicitly stated in the requirements document that was provided.

What Is EOIS?

The **Executive Office Intelligence System (EOIS)** is a bespoke digital assistant built specifically for the Group Vice President of Dangote Group. Think of it as an always-available, highly intelligent Chief of Staff — a system that reads and sorts through the GVP's emails, keeps track of meetings, manages tasks and deadlines, stores and analyses important documents, generates daily briefings, and even sends real-time alerts directly to the GVP's phone via WhatsApp when something urgent comes up.

To do all of this reliably and securely, EOIS relies on seven carefully chosen external technology platforms. Each one handles a specific job. This document explains what each service is, what it does in simple terms, and exactly how and where it fits into the EOIS system.

SECTION 01

Microsoft Entra ID

What Is It? Microsoft Entra ID — previously known as Azure Active Directory — is Microsoft's cloud-based identity and login management service. Every major organisation that uses Microsoft 365 (Outlook, Teams, SharePoint, etc.) already runs on Entra ID behind the scenes.

In Plain English: Entra ID is the "front door security guard" for EOIS. Instead of creating a completely separate username and password system just for this application, EOIS uses the organisation's existing Microsoft 365 login. When a user opens EOIS and clicks "Sign In with Microsoft", they are taken through Microsoft's own secure login page — including any two-factor authentication the organisation already requires. If their Microsoft account is deactivated (for example, when an employee leaves the company), they are automatically locked out of EOIS as well, with no extra steps needed.

Why EOIS Uses It

- **Single Sign-On (SSO):** Users log in once with their Microsoft account and gain access to EOIS. No extra passwords to remember or manage.
- **Automatic Email Access:** Because the GVP's Outlook inbox is part of Microsoft 365, once authenticated, EOIS can securely read and process incoming emails on his behalf —

automatically, in the background, every five minutes.

- **Automatic Calendar Access:** In the same way, EOIS reads the GVP's Outlook calendar every ten minutes to keep track of upcoming meetings, travel, and engagements.
- **Teams Notifications:** Entra ID credentials also allow EOIS to send system alerts into Microsoft Teams channels when important events or urgent items arise.
- **Inherited Security:** Any security policies the IT team has set on Entra ID — such as multi-factor authentication, login location restrictions, or device compliance — automatically apply to EOIS logins as well.

■ **Real-World Analogy:** Think of Entra ID like the key card system at the office. One key card opens the main building, the executive floor, and the private office — you do not need a separate key for every door. EOIS is simply one more "door" that uses the same key card system.

SECTION 02

OpenAI API

What Is It? OpenAI is the company behind the world-famous ChatGPT and GPT-4 artificial intelligence models. The OpenAI API gives software applications like EOIS direct access to these same powerful AI models, allowing the system to read, understand, summarise, and generate human-quality text.

In Plain English: OpenAI is the "brain" of EOIS. Without it, the system would just be a database — storing information but unable to understand it or act on it intelligently. With OpenAI, EOIS can read a complex, multi-page legal email and in seconds tell the GVP: "This is urgent, it requires a decision by Friday, and here is a suggested reply." It can also read hundreds of documents stored in the system and answer specific questions about them in natural, conversational language.

Why EOIS Uses It

- **Email Intelligence:** Every email that arrives in the GVP's inbox is automatically read and analysed by the OpenAI model. The AI assigns it a priority level (Urgent, High, Normal, or Low), identifies any tasks or decisions hidden inside the email, and can draft a suggested reply — all without human intervention.
- **Morning Briefing Generation:** Every morning at 6:00 AM, EOIS uses OpenAI to generate a personalised, structured briefing document for the GVP. It pulls in the day's calendar events, pending tasks, important unread emails, and outstanding decisions, then writes a coherent, readable summary — much like a Chief of Staff would prepare.

- **The AI Assistant (Chat):** The "AI Assistant" feature on the EOIS dashboard allows the GVP or his team to ask questions in plain English — for example, "What decisions are pending from last month's board meeting?" or "Summarise the Q3 financial report." OpenAI reads the relevant documents from the system and provides a clear, accurate answer.
- **Document Understanding:** When a new document is uploaded into EOIS (a board presentation, a financial report, a briefing note), OpenAI reads it, extracts the key information, and creates a searchable "fingerprint" of its contents so it can be found and referenced later.

■ **Real-World Analogy:** OpenAI is like having an exceptionally well-read, tireless analyst on the team who has read every document, every email, and every report stored in the system — and can answer any question about them instantly, at any time of day.

SECTION 03

Azure Database for PostgreSQL

What Is It? Azure Database for PostgreSQL is Microsoft's cloud-hosted version of PostgreSQL — one of the world's most trusted and widely-used database systems. A database is, at its most basic, an organised filing cabinet for digital information, designed to store, retrieve, and update data quickly and reliably.

In Plain English: The Azure Database is the "master filing cabinet" for EOIS. Every single piece of structured information the system handles — every email record, task, calendar event, uploaded document entry, pending decision, user account, notification, and meeting note — is stored here. When the GVP opens the dashboard and sees his upcoming meetings, those records are being read from this database in real time.

Why EOIS Uses It

- **Stores All Application Records:** The database holds 15+ categories of information including users, emails, tasks, calendar events, decisions, briefings, documents, commitments, meetings, and notifications. Nothing is lost — all data persists reliably even after system restarts.
- **Enables Filtering and Search:** When the GVP asks to see all "Urgent" emails from this week, or all tasks assigned to a specific department, the database engine can find and return exactly those records in milliseconds from potentially thousands of entries.
- **Keeps Everything Consistent:** If two people update the same record at the same time — for example, two members of the executive team both marking a task as complete — the database handles this safely, ensuring data remains accurate and consistent.
- **Secure and Backed Up:** As a managed Microsoft Azure service, the database is automatically backed up, monitored for security vulnerabilities, and updated — reducing the operational burden on the development team.

■ **Real-World Analogy:** Think of this as the executive office's central filing room — meticulously organised, always available, and containing every document, record, and note ever created. Any authorised staff member can walk in, find exactly what they need in seconds, and the room is always locked and guarded.

SECTION 04

Azure Blob Storage

What Is It? Azure Blob Storage is Microsoft's cloud-based service for storing large files — documents, images, videos, spreadsheets, and any other type of file — at virtually unlimited scale. "Blob" simply stands for Binary Large Object, which is technical language for "a file."

In Plain English: If the Azure Database is the "filing cabinet" for structured records, then Azure Blob Storage is the "physical document safe" where the actual files are kept. When the GVP's team uploads a board presentation PDF, a financial Excel spreadsheet, or a confidential briefing note into EOIS, the file itself goes into Azure Blob Storage. The database simply holds a note about the file — its name, category, and date — and a pointer to where the actual file lives in the storage safe.

Why EOIS Uses It

- **Document Storage:** Every file uploaded to the EOIS Document Intelligence section — PDFs, Word documents, Excel files, PowerPoint presentations — is securely stored in Azure Blob Storage inside an organised folder structure (Board Documents, Financial Reports, Briefing Notes, etc.).
- **File Retrieval on Demand:** When a user clicks to open or download a document from the EOIS interface, the system fetches it directly from Blob Storage and delivers it to the user's browser — fast and securely.
- **AI Processing Pipeline:** When a new document is uploaded, the AI analysis process downloads the file from Blob Storage, reads its text content, and passes it to OpenAI for understanding and indexing. Without reliable file storage, this pipeline cannot function.
- **Cost-Effective at Scale:** The GVP's office will accumulate hundreds or thousands of documents over time. Blob Storage is designed for this scale and costs a fraction of what storing files in a traditional database would cost.

■ **Real-World Analogy:** Blob Storage is like a highly secure, climate-controlled archive room in the office building. The filing cabinet (database) has index cards telling you exactly what is in the archive room and where to find it — but the actual physical documents themselves are locked away in the archive for safe-keeping.

SECTION 05

WhatsApp Business API

What Is It? The WhatsApp Business API is Meta's official service that allows organisations to send and receive WhatsApp messages programmatically — meaning a software system can send WhatsApp messages automatically, without a human physically typing them. This is the same WhatsApp platform used by billions of people every day, but accessed through a business-grade, secure interface.

In Plain English: WhatsApp is the GVP's fastest and most direct communication channel. No matter where in the world he is — in a meeting, travelling, or at dinner — his phone is with him. EOIS uses the WhatsApp Business API to send instant alerts directly to his WhatsApp when something truly urgent happens that cannot wait until he next checks the EOIS dashboard. Think of it as the emergency notification system.

Why EOIS Uses It

- **Urgent Email Alerts:** When the AI email triage system identifies an incoming email as "Urgent" — meaning it requires the GVP's immediate attention — EOIS can automatically send a WhatsApp notification to his phone with a summary of the email and the action required.
- **Critical Deadline Reminders:** If a decision deadline is approaching and the item has not yet been resolved, EOIS can dispatch an automated WhatsApp reminder to ensure nothing falls through the cracks.
- **Time-Sensitive Calendar Alerts:** For last-minute meeting changes, cancellations, or new urgent appointments, WhatsApp serves as a reliable real-time channel that does not depend on the GVP being logged into any system.
- **A Secondary Safety Net:** While EOIS has its own in-app notification system, WhatsApp provides a completely independent channel — ensuring that even if the GVP is not actively looking at the EOIS dashboard, critical information still reaches him instantly.

■ **Real-World Analogy:** The WhatsApp integration is like having a dedicated executive assistant who can tap the GVP on the shoulder — no matter where he is — and say: "Sir, this one cannot wait." All other notifications come through normal channels. WhatsApp is reserved for what truly matters right now.

SECTION 06

Azure Cache for Redis

What Is It? Azure Cache for Redis is a Microsoft-managed, ultra-fast data storage service that keeps frequently needed information in the computer's active memory (RAM) rather than on a hard disk. Redis is also widely used as a task queue — a system for organising and managing background jobs that need to happen automatically without disrupting the main application.

In Plain English: Redis serves two jobs in EOIS. First, it is a "speed booster" — it remembers answers to questions that get asked frequently, so the system does not have to go all the way to the database every single time. Second, it is a "work scheduling manager" — it maintains a queue of background jobs that EOIS needs to perform automatically (like checking for new emails, syncing the calendar, or generating the morning briefing), making sure each job runs at the right time and in the right order.

Why EOIS Uses It — As a Speed Booster (Cache)

- **Storing Pre-Built Answers:** When the morning briefing is generated at 6 AM, it takes a few seconds of AI processing to create. Once done, Redis stores the finished briefing temporarily. If the GVP opens EOIS at 7 AM and again at 9 AM, the system retrieves the cached briefing instantly rather than regenerating it from scratch each time.
- **Search Suggestions:** When users type in the search bar, EOIS shows instant suggestions. Redis stores the most common search results so they appear immediately without querying the database afresh every keystroke.
- **System Performance:** By caching popular data, Redis prevents the database from being overwhelmed with repetitive requests, keeping the entire application fast and responsive for all users simultaneously.

Why EOIS Uses It — As a Work Scheduler (Task Queue)

- **Email Polling (Every 5 minutes):** EOIS cannot leave the inbox checking to chance. Redis manages a recurring background task that instructs the system to check the GVP's Outlook inbox every five minutes, completely automatically.
- **Calendar Sync (Every 10 minutes):** Similarly, Redis schedules automatic calendar synchronisation every ten minutes to keep the GVP's schedule up to date in real time.
- **Document Sync (Every 30 minutes):** Any new documents added to the GVP's OneDrive are automatically detected and brought into the EOIS document repository every thirty minutes.
- **Morning Briefing Generation (Daily at 6 AM):** The daily AI briefing is queued as a scheduled Redis task, ensuring it is ready before the GVP arrives at the office each morning.

■ **Real-World Analogy:** Redis plays two roles simultaneously. As a cache, it is like a personal assistant's notepad — quick answers to common questions written down for instant reference. As a task queue, it is like the daily scheduling book — a master list of everything that needs to happen today, what time, and in what order, ticked off as each item is completed.

SECTION 07

Azure Container Apps

What Is It? Azure Container Apps is Microsoft's cloud hosting service for running modern applications packaged as "containers." A container is a self-contained, portable package that includes an application and everything it needs to run — its code, settings, and dependencies — bundled together so it runs identically in any environment.

In Plain English: Azure Container Apps is the "office building" that the entire EOIS system lives in and operates from. Just as a business needs a physical or virtual space from which to operate, EOIS needs a reliable, secure, always-online home in the cloud. Azure Container Apps provides that home — and crucially, it can automatically expand that space (add more capacity) when many people are using the system at once, and shrink it back down during quiet periods to save cost.

How EOIS Is Structured Inside Azure Container Apps

EOIS does not run as a single large application. It is split into three separate, independent components, each running as its own container:

Component	What It Does
API Server (eois-api)	The main "command centre" — handles all requests from the web browser, runs the logic
Web Frontend (eois-frontend)	The visual interface — the dashboard, communications hub, and all the screens the GVP
Background Worker (eois-worker)	The "always-busy" component running in the background — handles scheduled tasks like

Why EOIS Uses It

- **Always Online:** Azure Container Apps runs the EOIS system 24 hours a day, 7 days a week. It automatically restarts any component that fails, ensuring the GVP always has access to his system.
- **Automatic Scaling:** If the entire executive team logs in simultaneously first thing Monday morning, Azure automatically spins up additional capacity to handle the load. When demand drops, it scales back down — paying only for what is used.
- **Security and Secrets Management:** Sensitive credentials — such as the OpenAI API key, the database password, and the WhatsApp access token — are stored securely as Azure secrets and injected into the containers at runtime. They are never written into the application code.

- **Custom Domain and HTTPS:** Azure Container Apps manages secure HTTPS web traffic and custom domain routing (e.g., eoism.dangote.com), ensuring all data transmitted between the GVP's browser and the system is encrypted.

■ **Real-World Analogy:** Azure Container Apps is the office building with a smart reception desk. The building stays open 24/7. If more visitors arrive than expected, extra floors open up automatically. The building's security team (Azure) manages access, fire safety, and building maintenance — so the office occupants (EOIS team) can focus entirely on their work.

SECTION

Summary — All Seven Integrations at a Glance

Integration	Simple Description	Core Role in EOIS
Microsoft Entra ID	The organisation's login system	Authenticates all users; gives EOIS secure access to Outlook email and
OpenAI API	The AI brain (GPT-4o model)	Reads emails, generates briefings, powers the AI assistant, understands
Azure Database (PostgreSQL)	The master filing cabinet	Stores every record: emails, tasks, events, decisions, users, documents
Azure Blob Storage	The document safe	Stores all uploaded files: PDFs, Word docs, Excel sheets, presentations
WhatsApp Business API	The emergency notification line	Sends instant phone alerts for urgent emails, deadlines, and critical even
Azure Cache (Redis)	The speed booster & task scheduler	Caches frequent data for speed; manages all automated background job
Azure Container Apps	The office building in the cloud	Hosts and runs all EOIS components reliably, securely, and at scale 24/

Each of the seven services documented in this report was chosen not at the discretion of the development team, but as a direct result of the requirements set out in the project requirements document. Together, they form a cohesive, enterprise-grade technology foundation that is secure, scalable, and fit for the demands of an executive office at the level of the Dangote Group.

End of Document — EOIS Integration Reference — July 2026